

Deterministic Optimization

Further Development of Simplex Method

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Handling Degeneracy

Further Development of Simplex Method

Learning Objectives for this lesson

- Recognize the phenomenon of degeneracy and its impact on simplex method
- Discover algorithmic methods to deal with degeneracy in simplex method

Degeneracy in LP

- Degeneracy is a common phenomenon in linear programs appearing in real-world applications
- Degeneracy of a BFS refers to the fact that redundant active constraints exist at a BFS
- For standard form LP, this means some basic variables are zero
 - For example, a BFS $x = [0, 2, 3, 0, 0]^T$ where x_1, x_2, x_3 are basic variables, then this BFS is degenerate. There are 4 constraints active at x .

Impact of Degeneracy on Simplex Iterate

- Let's continue the example.
- Suppose at the start of the current iteration, the BFS is degenerate:
 - $x = [0, 2, 3, 0, 0]^T$, $x_B = [0, 2, 3]^T$, $x_N = [0, 0]^T$
 x_1, x_2, x_3 x_4, x_5
- Suppose the simplex method found a descent direction:
 - $d = [-1, -1, 2, 1, 0]^T$
 - That is, x_4 enters the basis
- By min-ratio test:
 - $\min \left\{ \frac{0}{-1}, \frac{2}{-1} \right\} = 0$, so x_1 exists the basis
- The new BFS:
 - $x = [0, 2, 3, 0, 0]^T$, $x_B = [2, 3, 0]^T$, $x_N = [0, 0]^T$
 x_2, x_3, x_4 x_1, x_5

The new BFS is the same point as the starting BFS! Simplex method Stayed at the same point. This can only happen to degenerate BFS.

Cycling of Simplex Method

- It is possible that the simplex method does not move away from a BFS solution due to degeneracy
- Simplex method can get stuck at a degenerate BFS
- All it does is from one basis to another and cycling back to the starting basis. Each BFS reached in the cycle is a new BFS with a new basis, but they all correspond to the same extreme point
- This phenomenon is called **cycling**. Simplex never terminates.

Ambiguities in Simplex Method

1. We start the simplex method with a basic feasible solution $\mathbf{x}$ and the associated basis consisting of the basis matrix $\mathbf{B} = [\mathbf{A}_{B(1)}, \dots, \mathbf{A}_{B(m)}]$, where $\mathbf{A}_{B(i)}$'s are the basic columns of $\mathbf{A}$.
2. Compute the reduced cost $\bar{c}_j = c_j - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_j$ for each nonbasic variable x_j . There are two possibilities:
 - (2.1) If all the reduced costs are nonnegative, the current BFS is optimal, and the algorithm terminates;
 - (2.2) Otherwise, select some j with $\bar{c}_j < 0$, so x_j enters the basis.
3. Compute $\mathbf{d}_B = -\mathbf{B}^{-1} \mathbf{A}_j$. There are two possibilities:
 - (3.1) If $\mathbf{d}_B \geq \mathbf{0}$, we know the optimal cost is $-\infty$, and the algorithm terminates.
 - (3.2) Otherwise, if some entry of $\mathbf{d}_B$ is negative, continue to the next step.
4. If some entry of $\mathbf{d}_B$ is negative, compute the stepsize θ^* by the following *min-ratio test*:

$$\theta^* = \min_{\{i=1, \dots, m \mid d_{B(i)} < 0\}} \frac{x_{B(i)}}{-d_{B(i)}}.$$

Suppose the index $B(l)$ achieves the minimum: $x_{B(l)}$ exits the basis.

5. Form the new basis matrix $\bar{\mathbf{B}}$ by replacing $\mathbf{A}_{B(l)}$ column with $\mathbf{A}_j$. The new BFS $\mathbf{y}$ has basic variable part $y_{B(i)} = x_{B(i)} + \theta^* d_{B(i)}$ and $y_j = \theta^*$. Start a new iteration.

How to select if multiple $\bar{c}_j < 0$?

How to select among multiple $B(l)$?

Pivoting Rules

- A pivot rule refers to a set of rules to unambiguously choose a nonbasic variable to enter basis and a basic variable to exit the basis.
- In particular, a pivot rule breaks ties when multiple nonbasic variables have negative reduced costs and multiple basic variables achieve the minimum ratio.
- There are many pivot rules such as always pick the most negative reduced cost or pick the first negative reduced cost found or break ties by random selection.
- Pivoting rule is important for the performance of simplex method.

Bland's Rule

- Bland's rule is a specific pivoting rule described below:
 - Choose the entering nonbasic variable x_j such that j is the **smallest index** with $\bar{c}_j < 0$
 - Choose the exiting basic variable x_i such that i is the **smallest index** that achieves the minimum ratio in the min-ratio test.
- Bland's rule is also called “the smallest index rule”.
- **Theorem:** If the simplex method uses the Bland's rule, then it always terminates with a finite number of steps, i.e. no cycling can happen.
- Bland's rule is discovered and proved by Robert Bland, a professor in Cornell University.

Summary

- We learned:
 - We learned the impact of degeneracy on simplex method and a pivoting rule to prevent cycling